



FACULTY OF ENGINEERING, SCIENCES AND TECHNOLOGY

DATABASE MANAGEMENT SYSTEMS LAB

FINAL PROJECT PROPOSAL

INSTRUCTOR: *SHOAIB AHMED*

GROUP MEMBERS:

- BURHAN ALI (67491)
- EMAN LIAQUAT (67404)
- SADAF (67400)

INDEX

<u>S.NO</u>	<u>CONTENT</u>	<u>PAGE NO.</u>
1	Introduction	3
2	Objectives	3
3	Scope	4-5
4	System Overview	5
5	Functional requirements	6
6	Non-functional requirements	6
7	New Innovation and Enhancements	6
8	Tools	7
9	Methodology	7
10	Expected Outcomes	8
11	Conclusion	8
12	References	8

PROJECT TITLE:

Smart QuickBite Restaurant Management and Online Food Ordering System *(Database Management System)*

INTRODUCTION:

Traditional food ordering systems are often time-consuming, manual, and prone to errors such as incorrect orders, delayed processing, and inefficient record management. With no support for sophisticated reporting, business analytics, transaction management, and database optimisation strategies, the majority of small-scale restaurant systems only concentrate on basic order placement and storage.

The Smart QuickBite Restaurant Management and Online Food Ordering System, a database-driven web application that lets users explore menu items, place multiple-item orders, choose payment options, and monitor order status, is suggested as a solution to these issues. The system demonstrates practical implementation of advanced Database Management System (DBMS) concepts including normalization up to Fourth Normal Form (4NF), joins, aggregate functions, subqueries, stored procedures, triggers, indexing, and transaction management. Through a centralised and scalable platform, the suggested solution guarantees effective data processing, enhanced data integrity, optimised query performance, and improved restaurant management.

OBJECTIVES:

The key objectives of this project are as follows:

- Create a web-based system for managing restaurants.
- Process food orders and payments automatically.
- Normalize up to 4NF.
- Improve uniformity and cut down on redundant data.
- Produce business reports that are analytical.
- Execute sophisticated SQL procedures.
- Make use of triggers and stored procedures.
- Use indexing to boost performance.
- Assure data integrity and transaction security.
- Ensure that users have a professional experience.

SCOPE:

The scope of this project includes the following Database Management System (DBMS) components:

1. Database Design:

- Creating an organised relational database for online ordering and restaurant administration.
- Identifying entities, properties, and relationships.
- Developing ERD, Database Schema, DFD, Use Case Diagram, Activity Diagram, Sequence Diagram, and System Architecture Diagram.
- Creating database structures that are normalised up to Fourth Normal Form (4NF).

2. Database Structure:

- Relational tables are created and relationships are defined.
- Using foreign and primary keys correctly.
- Making many-to-many relationship junction tables.
- Implementing constraints such as NOT NULL, UNIQUE, CHECK, and DEFAULT.
- Ensuring consistency and integrity of data.

3. Database Operations:

- Executing CRUD (Create, Read, Update, Delete).
- Running complex SQL queries.
- Employing FULL OUTER JOIN, LEFT JOIN, RIGHT JOIN, and INNER JOIN.
- Aggregate functions like COUNT, SUM, AVG, MIN, and MAX are implemented.
- Using GROUP BY and ORDER BY for reporting.
- Implementing subqueries for advanced data retrieval.
- Creating sales summary and business reports.

4. Database Concepts:

- Normalization (1NF, 2NF, 3NF, BCNF, and 4NF).
- Functional Dependencies.
- Referential Integrity.
- Aggregate functions
- Stored Procedures;
- Triggers, Indexing

- Transactions and Rollback.
- Techniques for Database Optimisation.

5. System Limitations:

- Database design and implementation are the system's main focus.
- There is no real-time delivery tracking.
- Online payment gateway integration is not implemented.
- The availability of menu items is the only aspect of inventory management.
- This project does not include the creation of mobile applications.

SYSTEM OVERVIEW:

A database-driven online application, the proposed Smart QuickBite Restaurant Management and Online Food Ordering System is intended to effectively manage restaurant operations. The system consists of the following main components:

1. Customer (User) Component:

This component allows customers to register, browse menu items, place orders, view order history, select payment methods, and track order status. The database securely stores customer data.

2. Menu Item Component:

This component manages food categories and menu items. Food names, categories, descriptions, prices, and availability status are all stored there. Menu items can be added, updated, and removed by administrators as needed.

3. Order Component:

This component handles the complete ordering process. Orders involving more than one food item can be placed by customers. The system updates order status, manages payment information, stores order details, and computes total costs automatically. Order processing is dependable and consistent thanks to transaction management.

4. Admin Component:

This component is responsible for managing the entire system. Customers, menu items, orders, payments, and reports can all be managed by administrators. The dashboard offers analytical data produced by aggregate functions and SQL queries, including total customers, total orders, revenue figures, top-selling food products, and sales reports.

FUNCTIONAL CONDITIONS:

- Management and registration of customers.
- Perusing the menu and choosing meals.
- Ordering more than one item.
- Processing of payments.
- Tracking the status of orders.
- Creation of reports.
- Management of dashboards.
- Filtering and searching.

NON-FUNCTIONAL CONDITIONS:

- Scalability and Security
- Dependability
- Usability and Performance
- Sustainability

NEW INNOVATIONS AND ENHANCEMENTS:

The suggested technique offers a number of improvements over conventional meal ordering systems:

- Several foods in one order.
- Payment management that is integrated.
- Order history tracking for customers.
- Order status management that is dynamic.
- Revenue is calculated automatically.
- Dashboard for advanced reporting and analytics.
- Analysis of popular foods.
- Stored protocols for database operations that are automated.
- Triggers for recording activities.
- Indexing to improve queries.
- Secure order processing using transaction management.
- Up to 4NF normalisation.
- Features for searching and filtering.
- Racking sales and revenue performance.

TOOLS:

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Node.js
- Express.js

Database:

- Microsoft SQL Server

Tools:

- Visual Studio Code
- SQL Server Management Studio (SSMS)
- GitHub

METHODOLOGY:

The project will be implemented through the following systematic steps:

- System planning and requirement analysis.
- ERD and database schema design.
- Making relational database tables with primary and foreign keys.
- Determining entities, attributes, and relationships.
- Using normalization methods up to 4NF.
- Applying linkages and restrictions.
- Inserting sample data (100+ records).
- Creating front-end user interfaces with JavaScript, HTML, and CSS.
- Implementing backend functionality using Node.js and Express.js.
- Performing CRUD operations.
- Implementing aggregate functions and reporting queries.
- Using subqueries and JOIN operations.
- Developing triggers and stored procedures.
- Putting rollback and transaction systems in place.
- Verifying database operations and testing every module.
- Using GitHub to manage project versions.

EXPECTED OUTCOMES:

- A database system for food ordering and restaurant management that is completely operational.
- Efficient storage and retrieval of restaurant data.
- Normalization reduces redundant data.
- Better consistency and integrity of data.
- Indexing allows for faster query execution.
- Automated processes with stored procedures and triggers.
- Business analytics and advanced reporting.
- Putting sophisticated DBMS ideas into practice.
- A better comprehension of practical database applications.

CONCLUSION:

The Smart QuickBite Restaurant Management and Online Food Ordering System successfully illustrates how sophisticated Database Management System (DBMS) concepts can be applied in a real-world restaurant setting. The system guarantees data quality, consistency, security, and dependability while offering an organised and effective way to handle clients, menu items, orders, payments, and administrative tasks.

Important DBMS concepts have been successfully implemented through this project, including entity relationship modelling, normalisation up to Fourth Normal Form (4NF), primary and foreign key relationships, CRUD operations, aggregate functions, joins, subqueries, stored procedures, triggers, indexing, and transaction management. Through reporting and analytics, these ideas assist minimise redundant data, enhance query performance, preserve data integrity, and facilitate effective decision-making.

Additionally, the project improves practical knowledge of advanced SQL implementation, web-based database integration, and database design. Overall, the proposed system highlights how modern database technologies can be utilized to develop scalable, optimized, and well-organized solutions for restaurant management and online food ordering applications.

REFERENCES:

- Korth, H. F., Silberschatz, A., & Sudarshan, S. *Database System Concepts*.
- W3Schools SQL Tutorial (www.w3schools.com/sql)
- Microsoft SQL Server Documentation.